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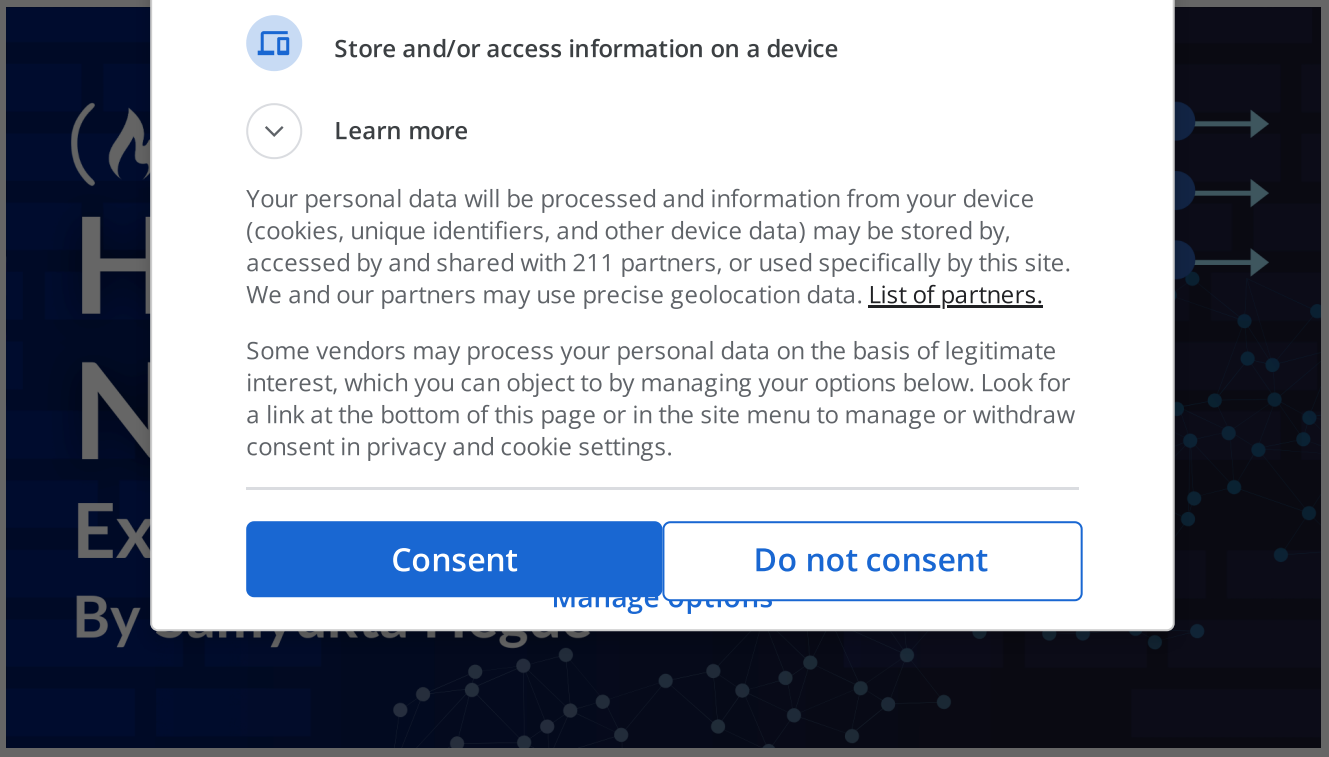
JANUARY 8, 2026 / #DEEP LEARNING

How Neural Networks Work – Explained Using the Str ax + b



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complex neural network starts with a fundamental question, “How does the output change when the input changes?“

A straight line equation `y = ax+b` answers it in the simplest way

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- Prerequisites
- y=ax+b
- Linear Regression
- Linear Classification
- Comparison
- Key Additions to Help Build Deep Neural Networks

Prerequisites

- A basic understanding of linear algebra, particularly `y=ax+b`.
- General idea about linear regression and classification.

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Ms. Poly is a math teacher who wants to formulate a study plan for her students to excel in an upcoming final exam. For simplicity, she creates a rule of thumb using only one factor: the number of hours studied per week. It has a direct impact on the marks scored by a student.

Before beginning, she makes certain assumptions:

- For every hour a student studies, an additional 3 marks can be scored.

She then comes up with the following equation based on her ideas: `y = 3x+30`

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In the above graph, she plots the points based on the results of the equation. As expected, it is a straight line. If she needs the marks scored for `9` hours of study, she can get it by just substituting `x=9` in `y=3x+30`. Note that the data (`x` and `y`) are just based on her hunch and aren't real.

her students and collects information on how many hours they study math per week. She creates a table with two columns: number of hours studied (`x`) per week and marks scored (`y`). She tries her old formula `y=3x+30`, but it doesn't seem to work. Thus, she doesn't have any sensible equation describing the relation bet

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new input data. Now, let's dive straight into the example to understand the concept better.

Ms. Poly is determined to arrive at a solution. She plots the collected data on a graph to get a better picture.

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Here, an expression is formulated to ascertain the change in output when the input changes. It looks like Ms. Poly is thinking like a data scientist. She has in fact modelled a very simple neural network for regression. The equation `y=4x+40` can be considered as the only neuron (processing unit) within it. She's adjusted the parameters `a` (weight) and `b` (bias) to arrive at the final formula which covers most of the points (thus minimizing the loss).

```
y => Marks scored.
x => Number of hours studied.
a=4 => Increase in marks for every hour studied
b=40 => Minimum marks
```

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defines the boundary condition for pass and fail.

If marks scored>=50, pass

If marks scored<50, fail

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neural network for classification. The equation x=3 is the only neuron within it. It can be considered to be having two parts as explained below.

and `b` (bias) to arrive at the following formula: `z = x-3`

```
z => Intermediate Value.
x => Number of hours studied.
a=1 => Influence of the number of hours studied on the
b=-3 => Minimum number of hours to study to pass the e
```

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classification used by Ms. Poly. **Regression** helps in predicting a value while **Classification** helps in decision making. Let's draw a small table to summarize the differences.

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Image Classification: An output label is produced based on the input images.

Text Translation: An English sentence can be given as an input to be translated to say, Spanish.

She probably should have to use a deep neural network if both data and task were complex. That presents another question: **How does one build a deep neural network?**

We will explore it further by extending the same example to a more realistic version.

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- Difficulty level of topics in math
- Pattern of the exam, and so on.

All the above neither act independently nor do they have a simple linear relation with the marks scored. So, she has to solve this problem by stacking the contributing factors one above the other in layers and also adding the element of non linearity. Let's take a look at each in detail.

Burnout leads to lower score whereas good sleep increases score. But burnout can be reduced if the student is well rested. So, the impact on the final score when these two factors interact should be taken into account. This is possible only when the system solves it in layers. The first layer can deal with how they independently influence the score, the next layer can explore the interaction be

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Modelling a Deep Neural Network

Ms. Poly should start the work on modelling a deep neural network by following the steps mentioned below:

process of modelling:

- What are the input features?
- What are the output features?
- What type of problem is it (regression/classification)?

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of the inputs. The general formula is `z=Wx+b`.

```
z: Intermediate value or Pre-activation
W: Weight matrix which consists of values corresponding to the i
each input feature
x: Matrix consisting of input features, [x1, x2, x3, x4, x5]
b: Bias which represents the initial assumptions of the teacher(
```

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examples. We can consider the following combinations assuming we have two neurons in this layer:

Neuron 1: It can focus on study hours, burnout, and rest, with other features contributing less significantly.

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activation function is applied to the intermediate value and can be expressed as:

a = g(z)

For example: `ReLU` is an activation function which outputs `z` only if `z` is positive, else `0`.

y = ReLU(z)=max(0,z)

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are values obtained from the previous layer. Likewise multiple layers can be stacked one over the other based on the requirements. In this example, for representation, we have taken two hidden layers with two neurons each. The number of layers and neurons can vary based on requirements.

Poly can decide what she wants for output: predict the marks scored by a student or predict if the student passes/fails the exam. If she wants the final marks scored, she just has to apply linear transformation in the neuron in the final layer to produce the output. If she wants pass/fail status, she has to apply both linear and non-linear transformations to achieve the desired results.

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The next steps are:

Training the model: The network is trained in the following way:

- Random weights and biases are assigned to the linear transformation portions of the network.

- If there are gaps between the actual result and the predicted result, corrections are made in weights and biases (this step is similar to what was done in linear regression and classification).
- The steps above are repeated until the results improve.

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